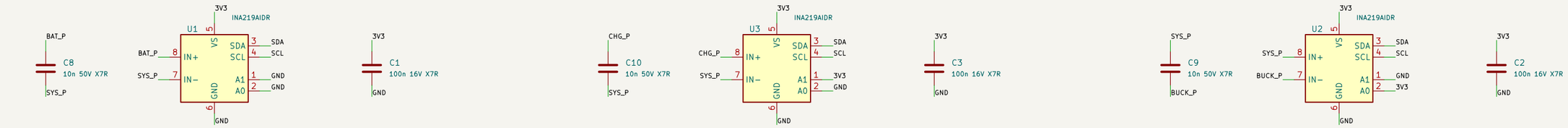
01 / PROTECTED DC POWER PATHS

BAT -> R1 -> SYS -> R2 -> BUCK. Charger -> R3 -> SYS. All returns use protected pack P-.



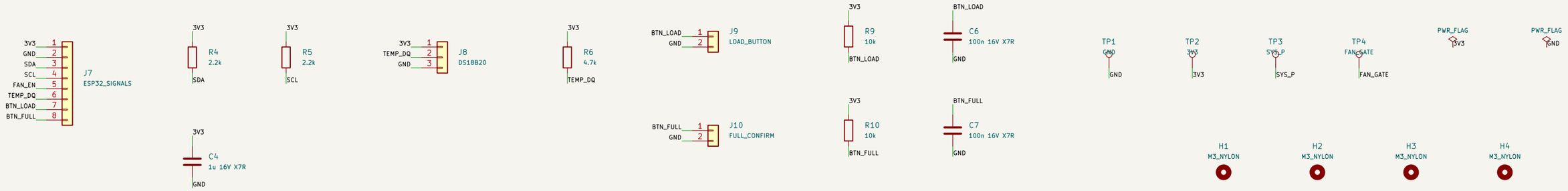
J1 requires a DC fuse near the battery AND a manual disconnect. J2: one matched charger only, <= 1 A; never connect a raw solar panel.

02 / THREE 3.3 V CURRENT MONITORS — SOIC PINOUT



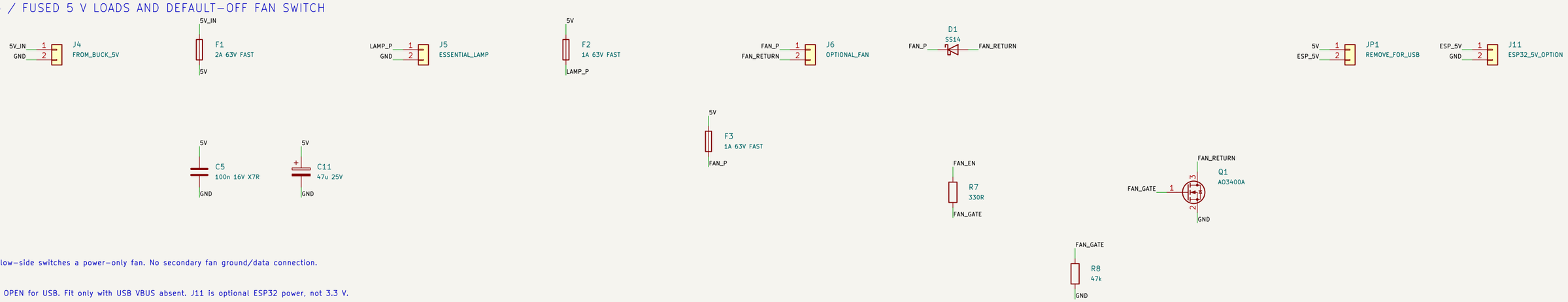
U1 0x40 battery / U2 0x41 buck input / U3 0x44 charger output. R1–R3 0.1 ohm, 1 W. Separate sense takeoffs at shunt pads.

03 / ESP32 SIGNAL HARNESS, TEMPERATURE AND LOCAL BUTTONS



J7 pins: 1=3V3, 2=GND, 3=GPIO21, 4=GPIO22, 5=GPIO25, 6=GPIO4, 7=GPIO27, 8=GPIO26. Keep cable <= 100 mm.

04 / FUSED 5 V LOADS AND DEFAULT-OFF FAN SWITCH



Q1 low-side switches a power-only fan. No secondary fan ground/data connection.

JP1 OPEN for USB. Fit only with USB VBUS absent. J11 is optional ESP32 power, not 3.3 V.

TARGET: 2 A battery path. 5 V continuous <= 1.5 A total; lamp and fan <= 0.6 A each. Include ESP32 consumption.

Prototype fabrication package: ERC/DRC are design checks; calibration, thermal and fault tests remain required.

Protected 4S LiFePO4 / 14.6 V / 2 A
Engineering prototype — hardware validation pending
Final-year electronics project

Sheet: /
File: wattsense-main.kicad_sch

Title: Wattsense main controller

Size: A3	Date: 2026-09-28	**Rev: B**
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